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Research and Implementation of Machine Learning Algorithms in Tourism Recommendation

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***Abstract*- The role of travel recommendation in the tourism industry is very important, but there is the problem that travel recommendation reviews are not true. Ordinary methods cannot solve the problem of travel recommendation in the tourism industry, and the recommendation is unreasonable. Therefore, this paper proposes a machine learning algorithm to conduct real travel recommendation analysis. First of all, the computer is used to collect and analyze the tourism review data, and the indicators are divided according to the requirements of tourism recommendation to reduce it Distractions in travel recommendations. Then, the computer analyzes the results of tourism recommendation in the tourism industry, forms a tourism recommendation plan, and recommends tourism The results were comprehensively analyzed. MATLAB simulation shows that under certain evaluation criteria, machine learning algorithms can make tourism recommendations accurate to the tourism industry The authenticity of travel recommendations is better than ordinary methods.**

Keywords- computer; machine learning algorithms; tourism industry; Travel recommendation

I. INTRODUCTION

With the continuous development of machine learning technology, its application in travel recommendation systems is becoming more and more extensive. The traditional travel recommendation system mainly achieves recommendation through user ratings, search records[1], browsing history, etc., but the shortcomings of these methods are becoming more and more obvious, such as data sparsity, cold start problems and low recommendation effect[2]. Therefore, the fusion of machine learning algorithms has become one of the key means to improve the quality of travel recommendations. This paper analyzes the influence and role of machine learning algorithms in the travel recommendation system from the perspective of their application[3].

A. Overview of the travel recommendation system

With the rapid development of Internet technology, the tourism industry has also undergone tremendous changes, and the transformation from offline tourism to online tourism has further promoted the development of travel recommendation systems[4]. The tourism recommendation system is an algorithm system that can recommend tourism products according to user needs, which can recommend tourism products for users according to their historical behavior, hobbies, social networks and other information, and provide users with personalized services[5]. Its main application scenarios include online ticket booking, travel guides, hotel reservations, etc.

At present, the travel recommendation system is mainly divided into two types: content-based recommendation and collaborative filtering-based recommendation. Content-based recommendations mainly recommend similar tourism products for users based on their historical behavior, evaluations and other information[6]; Recommendations based on collaborative filtering mainly recommend travel products similar to users' interests based on their historical ratings, browsing records and other information[7].

However, the traditional recommendation system has some problems, such as data sparsity, cold start problem and low recommendation effect, etc., in order to solve these problems[8], the introduction of machine learning algorithms has become one of the key means to improve the quality of travel recommendation[9].

B. Application of machine learning algorithms in travel recommendation systems

Machine learning algorithms are a technology based on data and statistical methods that automatically obtain patterns and patterns to achieve predictions and decisions[10]. In the travel recommendation system, machine learning algorithms are mainly used in the following aspects[11].

1) Natural language processing

Natural language processing is a technology that enables computers to understand natural language. In the travel recommendation system[12], natural language processing can be used to process text information such as reviews and ratings, and extract the user's emotional information to better judge the user's interests and needs[13]. Among them, sentiment analysis technology can be used to identify users' evaluation of tourism products, while text classification technology can be used to judge users' attitudes towards tourism products (such as likes, dislikes, etc.)[14].

2) Feature extraction

Feature extraction refers to the process of transforming raw data into meaningful feature vectors. In the tourism recommendation system, feature extraction can be used to extract features such as users' historical behavior and evaluation[15], as well as the description and evaluation of tourism products. For example, information such as users' historical ratings, click records, and comments can be converted into user interest vectors to better describe users' interests, hobbies and behavior habits; At the same time, the description, evaluation and other information of tourism products can be converted into feature vectors to better express the characteristics and selling points of tourism products[16].

3) Modeling and forecasting

Modeling and prediction are the core parts of machine learning algorithms, mainly including classification, clustering, regression and other techniques. In the travel recommendation system, user models and tourism product models can be built through these technologies to better predict users' interests and recommend travel products[17].

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For example, clustering techniques can be used to divide users into different groups to better understand their needs and behaviors; At the same time, classification techniques can be used to classify tourism products into different categories to better understand and describe the characteristics of tourism products[18].

4) Model optimization

In order to improve the recommendation effect of machine learning algorithms, the model needs to be optimized. Among them, the most common method is to avoid model overfitting and underfitting through cross-validation and regularization. In addition, methods such as deep learning and reinforcement learning can be used to further improve the recommendation effect.

C. The impact of machine learning algorithms on travel recommendation systems

The application of machine learning algorithms in the travel recommendation system can help the system better understand user needs, improve the recommendation effect and user experience, and thus have a positive impact on the travel recommendation system. The following is an analysis of its impact in three areas.

1) Improve referral performance

Machine learning algorithms can improve the quality of recommendations by better mining user interests and needs through techniques such as natural language processing, feature extraction, modeling, and prediction. For example, by converting information such as users' historical behaviors and evaluations into feature vectors, users' behavioral habits and interests can be better described; At the same time, by building user models and tourism product models, we can better understand user needs and characteristics of tourism products, and improve the personalization of recommendations to users.

2) Resolve data sparsity and cold start issues

Traditional recommendation systems are prone to data sparsity and cold start problems, that is, when the amount of data for users and travel products is small, the system cannot build effective models. Machine learning algorithms can solve data sparsity and cold start problems by extracting and modeling data to improve the accuracy and efficiency of recommendations.

3) Improve the user experience

Machine learning algorithms can improve user experience by better understanding user needs and feedback through techniques such as natural language processing and sentiment analysis. For example, sentiment analysis can be used to identify users' reviews of travel products to improve recommendation strategies; At the same time, natural language processing can be used to identify problems and suggestions raised by users to improve service quality and user satisfaction.

In summary, the application of machine learning algorithms in the travel recommendation system can effectively improve the recommendation effect, solve the problem of data sparsity and cold start, and improve the user experience. However, it should also be noted that the application of machine learning algorithms needs to combine actual business scenarios and user needs, and reasonably select algorithms and parameter settings to achieve the best recommendation effect.

II. Related Concepts

A. Mathematical description of a machine learning algorithm

As consumers' demand for travel experience continues to escalate, it is necessary to continuously optimize tourism products and improve the quality of travel experience. This can be achieved through the in-depth development of tourism resources, the improvement of the differentiation of tourism products, and the increase of cultural connotation of tourism products.

Hypothesis 1: The travel recommendation requirement is a_i, the travel recommendation plan is set_i, the satisfaction of the travel recommendation plan is x_i, and the travel recommendation plan judgment function is $Q(a_i \approx 0)$, As shown in Equation (1).

$$Q(a_i x_i) = \sum_{i=1}^{a} x_i^2 + \sqrt{3} + \sqrt{a_i^2 - x_i^2} \cdot \iint Q \tag{1}$$

B. Selection of travel review programs

Hypothesis 2: The tourism industry function is $b(a_i)$, and the weight coefficient is y_i, then the tourism recommendation requirements for unqualified tourism industries are shown in Equation (2).

$$b(a_i) = \bigcap_{i=1} x_i \cdot {}^{3}\!/_{b} + \sqrt{a_i^2 + y_i - \sqrt{3}} \cdot \bigcup_i^b a \tag{2}$$

C. Analysis of travel recommendation programs

Improve the quality of tourism services and enhance the satisfaction of tourists. This can be achieved by improving the quality of service personnel, strengthening the construction of tourist facilities, and optimizing tourist routes.

Integrated development of tourism industry: Promote the integrated development of tourism and related industries to form a more complete tourism industry chain. This can include integration with the cultural industry, integration with agriculture, integration with the sports industry, etc. and the specific scheme selection is shown in Figure 1.

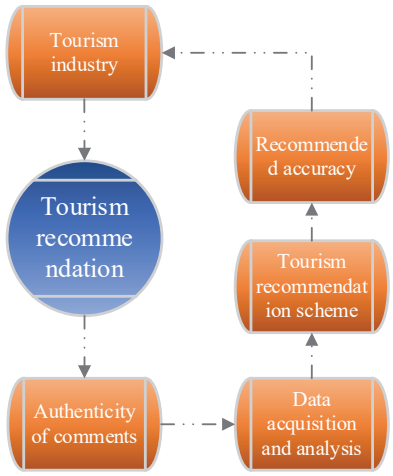

Fig. 1. The results of the selection of travel review programs

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III. Optimization Strategies for the Tourism Industry

Innovation-driven development: Promote the transformation and upgrading of the tourism industry through scientific and technological innovation, model innovation, management innovation, etc., and improve the competitiveness and sustainable development of the tourism industry.

Strengthen policy support: The government can introduce relevant policies to support the development of the tourism industry, such as tax incentives, financial support, talent training, etc. At the same time, the government can also strengthen the supervision of the tourism industry, regulate the market order, and protect the rights and interests of consumers.

Pay attention to environmental protection and sustainable development: The development of tourism should pay attention to environmental protection and sustainable development, protect tourism resources, and reduce the impact on the environment. At the same time, we should promote the development of green tourism, improve tourists' awareness of environmental protection, and realize the harmonious development of tourism and the environment.

IV. Practical Examples of the Tourism Industry

A. *Introduction of travel recommendations*

It is necessary to integrate with various resources in tourism and put forward corresponding strategies and methods, as follows.

TABLE I. Travel Recommendation Requirements

Scope of application	grade	accuracy	Travel reviews
reception	standard	89.34	87.33
	Higher	83.44	85.83
traffic	standard	84.42	86.12
	Higher	84.65	85.59
Recreation	standard	87.62	85.76
	Higher	82.69	85.52

The tour recommendation process in Table I. is shown in Figure 2.

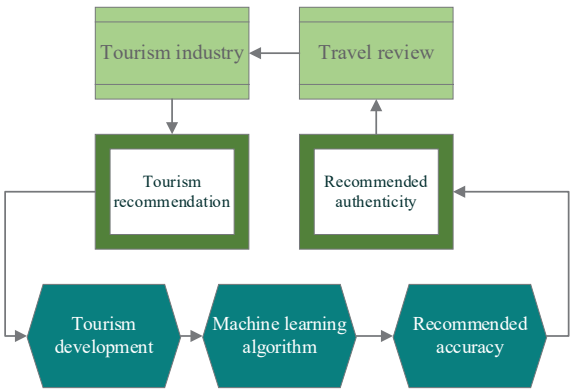

Fig. 2. The analysis process of the tourism industry

Tourism is one of the fast-growing industries worldwide, and it involves many fields and aspects, including hospitality, catering, transportation, culture, environment, etc. With the improvement of people's living standards and the increase of tourism consumption, the development potential of tourism is huge. However, the tourism industry also faces many challenges, such as fierce market competition, uneven service quality, and over-exploitation of tourism resources. Therefore, in order to improve the development level and competitiveness of the tourism industry, it is necessary to develop and implement optimization strategies.

B. *The situation of the tourism industry*

In-depth development of tourism resources: Different development strategies are adopted for different types of tourism resources. For natural scenery resources, attention should be paid to ecological protection and sustainable development, for historical and cultural resources, attention should be paid to cultural inheritance and historical reproduction, and for urban tourism resources, attention should be paid to urban cultural characteristics and tourism experience.

TABLE II. The Overall Picture of the Travel Review Program

category	Satisfaction	Analysis rate
reception	87.54	86.43
traffic	88.81	86.80
Recreation	85.82	89.64
mean	87.25	87.32
X^6	91.54	87.25
	P=1.76	

C. *Travel reviews and stability of travel recommendations*

We develop tourism resources and realize the integration of tourism resources.

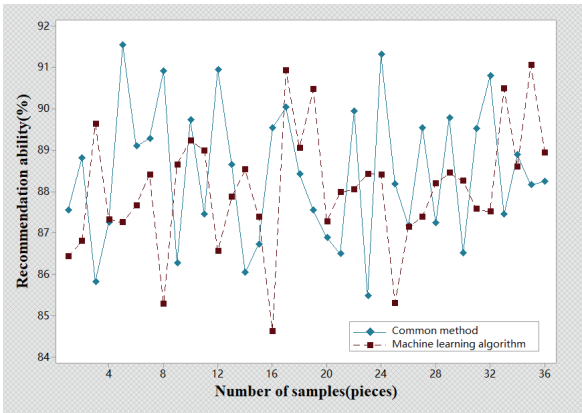

Fig. 3. Travel reviews with different algorithms

The machine learning algorithm outperforms ordinary methods in terms of travel reviews, with higher ratings and lower error rates, suggesting the superior stability of travel recommendations generated by machine learning algorithms compared to ordinary methods. The average travel recommendation scheme of the three algorithms mentioned above can be found in Table III.

TABLE III. Comparison of Travel Recommendation Accuracy by Different Methods

algorithm	Travel reviews	Magnitude of change	error
Machine learning algorithms	94.25	92.29	93.91
Normal method	92.21	91.39	92.88
P	92.28	91.19	93.01

Table III shows that the general method has shortcomings in the accuracy of tourism reviews in the tourism industry, and the tourism industry has undergone significant changes, and the error rate is high. The general results of machine

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learning algorithms have higher travel reviews than ordinary methods.

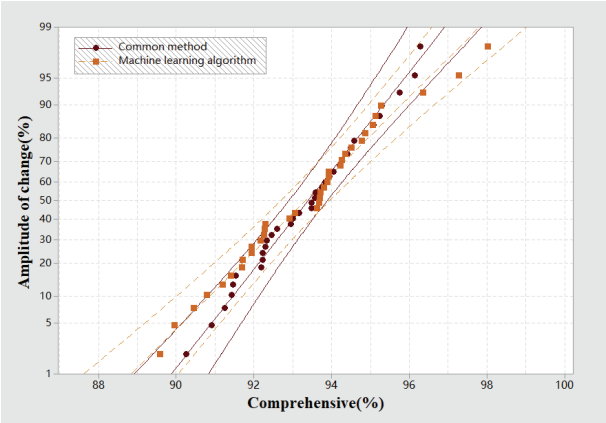

Fig. 4. Travel-recommended travel reviews by machine learning algorithms

V. CONCLUSION

Implement modern management concepts and methods to improve the management level and efficiency of the tourism industry. For example, the introduction of lean management, total quality management and other management methods. Seventh, strengthen policy support, the government can introduce relevant policies to support the development of tourism, such as tax incentives, financial support, personnel training, etc., at the same time, the government can also strengthen the supervision of the tourism industry, standardize market order, protect consumer rights and interests, eighth, pay attention to environmental protection and sustainable development, the development of tourism should pay attention to environmental protection and sustainable development, protect tourism resources, reduce the impact on the environment, and should promote the development of green tourism. Research shows that machine learning algorithms can improve the accuracy and authenticity of travel recommendations on the tourism industry Make general tour recommendations. However, in the process of machine learning algorithms, too much attention is paid to the analysis of travel recommendations, resulting in irrationality in the selection of tourism recommendation indicators.

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